

Discussion 3

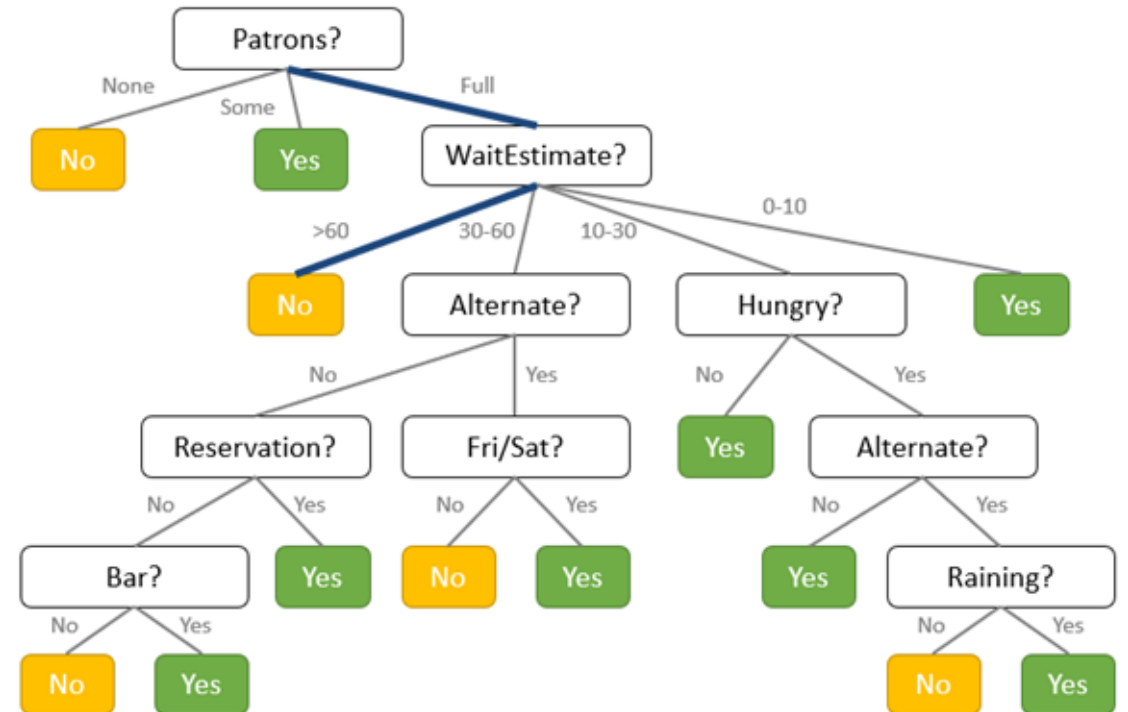
Decision Trees and Ensembles

Logistics

- Homework 1 has been released!

Decision Trees

- Model for classification problems
- Inputs are attribute values, output is a decision label
- Decision trees are comprised of a sequence of tests
 - **Internal nodes** test attribute values
 - From internal nodes, **branches** are labeled with possible values of an attribute
 - **Leaf nodes** indicate the return value (classification label)



Example DT from Lecture

Construction of Decision Trees

- Intuition:
 - Choose the attributes that would give us the most information with a single question
 - Metric that allows calculating information?
 - Entropy!

Entropy

- For a random variable V , we define the entropy H based on the possible values v_k that V can take:

$$H(V) = -\sum_k P(v_k) \log_2 P(v_k)$$

- For a boolean random variable that is true with probability q , we use different entropy notation for simplicity:

$$B(q) = -(q \log_2 q + (1 - q) \log_2 (1 - q))$$

Note that though the notation is different, the computation is the same as for regular random variables!

Entropy trivia

- Why log - base 2?
 - $\log_2$ counts how many times you can halve uncertainty.
- Entropy then, is **average number of binary distinctions needed to learn the outcome.**
- Fun fact, log base 2 are called “Shannons”



Constructing Decision Trees

- While constructing the decision tree, we will select the *best* attribute to test for; but what does “best” mean?
- We need a metric for computing how much information we gain from the attribute toward deciding the classification label
- We use ***entropy* (H)**, a measure of uncertainty about a random variable in number of bits

$$H(V) = -\sum_k P(v_k) \log_2 P(v_k)$$

$$B(q) = -(q \log_2 q + (1 - q) \log_2 (1 - q))$$

Entropy Practice

We want to determine if a frog is poisonous from the given observations.

What is the entropy of the target Poisonous variable?

Observation	Size	Color	Spots	Poisonous
1	Small	Blue	No	Yes
2	Small	Blue	Yes	Yes
3	Big	Blue	No	No
4	Big	Green	No	No
5	Small	Green	Yes	Yes
6	Small	Blue	No	Yes

Selecting Attributes

- Once we know the entropy of the target/goal variable, we can compute how much information each attribute gives us
- Consider an attribute A with d distinct values, where each value $1 \leq k \leq d$ has p_k positive and n_k negative examples
- We can compute the ***remainder*** of the data when split on attribute A

$$\text{Remainder}(A) = \sum_{k=1}^d \frac{p_k + n_k}{p + n} B \left(\frac{p_k}{p_k + n_k} \right)$$

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- We want to split on the attribute with the highest **information gain**

$$\text{Gain}(A) = B \left(\frac{p}{p+n} \right) - \text{Remainder}(A)$$

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$$\text{Gain}(A) = B\left(\frac{p}{p+n}\right) - \text{Remainder}(A)$$

Entropy of the **remaining** examples.
This will change after each branch!

Information Gain Example - SOLUTION

$$B(q) = -(q \log_2 q + (1 - q) \log_2 (1 - q))$$

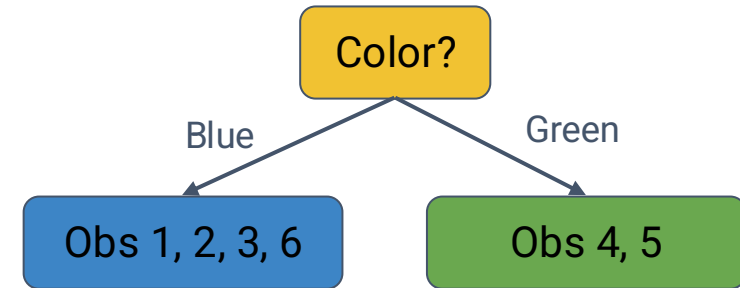
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Suppose we split on color first.

What attribute should we split on next for the Blue examples?

First, compute entropy of remaining examples. Binary variable with $q=3/4$



Observation	Size	Color	Spots	Poisonous
1	Small	Blue	No	Yes
2	Small	Blue	Yes	Yes
3	Big	Blue	No	No
4	Big	Green	No	No
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Information Gain Example - SOLUTION

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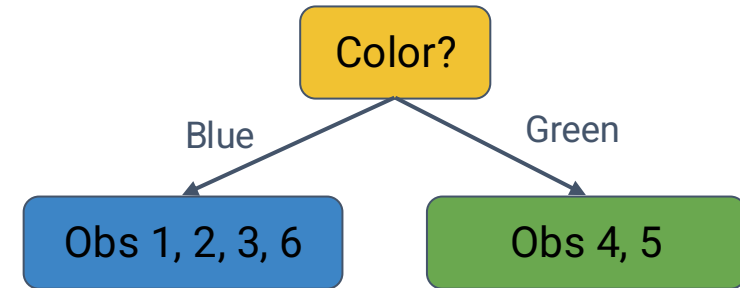
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Suppose we split on color first.

What attribute should we split on next for the Blue examples?

First, compute entropy of remaining examples. Since 3/4 examples are Yes, we treat it as a binary variable with $q=3/4$

$$B\left(\frac{3}{4}\right) = -\left(\frac{3}{4}\right) \log_2 \left(\frac{3}{4}\right) - \left(\frac{1}{4}\right) \log_2 \left(\frac{1}{4}\right) = 0.81125$$



Observation	Size	Color	Spots	Poisonous
1	Small	Blue	No	Yes
2	Small	Blue	Yes	Yes
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Information Gain Example - SOLUTION

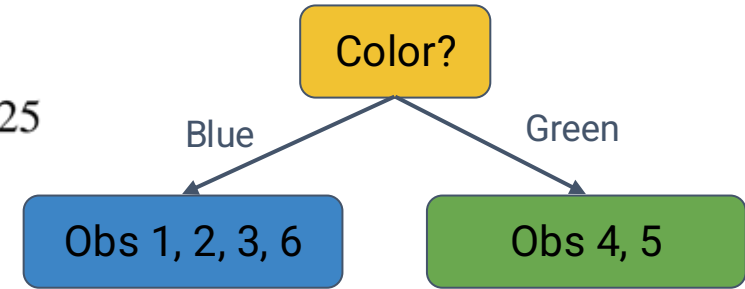
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Suppose we split on color first.

$$B\left(\frac{3}{4}\right) = 0.81125$$



What attribute should we split on next for the Blue examples?

Compute the gain of splitting on **Spots**

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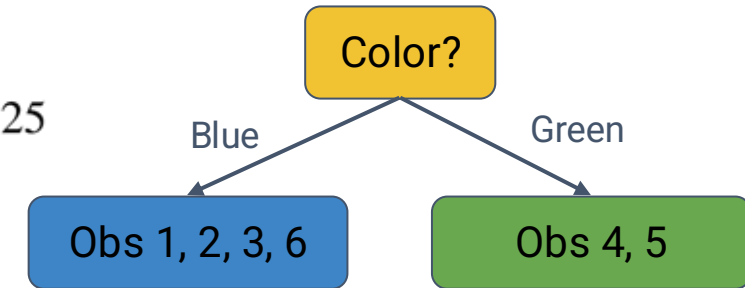
Compute the gain of splitting on **Spots**

Let's say k=1 is Spots=Yes, k=2 is Spots=No.

$p_{k=1} = 1$ (No. of poisonous=Yes is 1)

$n_{k=1} = 0$ (No. of poisonous=No is 0) and so on..

$$B\left(\frac{3}{4}\right) = 0.81125$$



Observation	Size	Color	Spots	Poisonous
1	Small	Blue	No	Yes
2	Small	Blue	Yes	Yes
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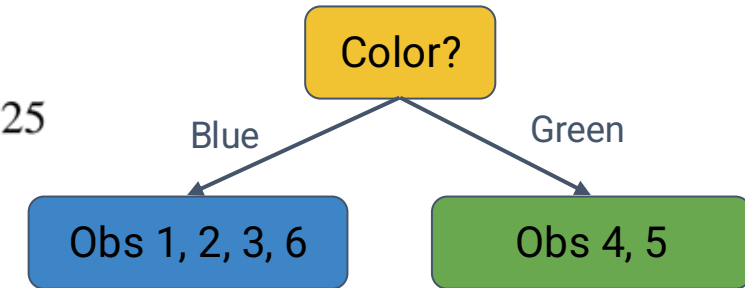
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Suppose we split on color first.

$$B\left(\frac{3}{4}\right) = 0.81125$$



What attribute should we split on next for the Blue examples?

Compute the gain of splitting on **Spots**

$$\text{Remainder}(\text{Spots}) = \left(\frac{1}{4}\right)B\left(\frac{1}{1}\right) + \left(\frac{3}{4}\right)B\left(\frac{2}{3}\right)$$

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1	Small	Blue	No	Yes
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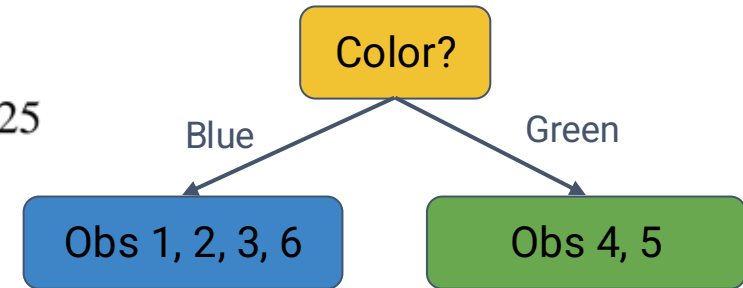
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Remaining
examples where
Spots=Yes

Entropy of
examples where
Spots=Yes

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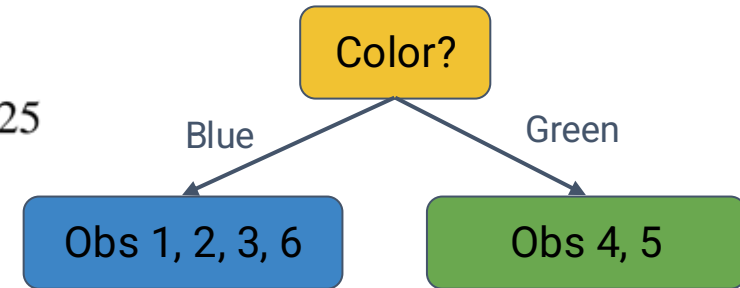
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Spots=No

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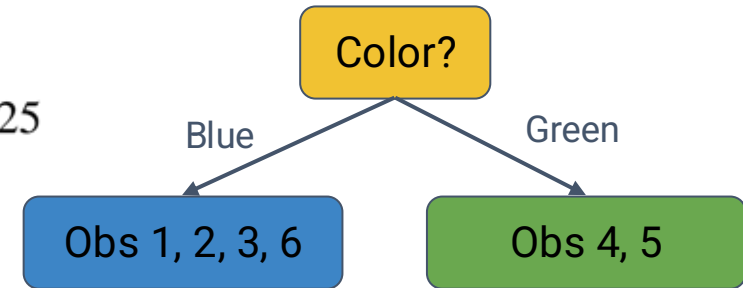
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$$\text{Remainder}(\text{Spots}) = 0 + 0.68875 = 0.68875$$

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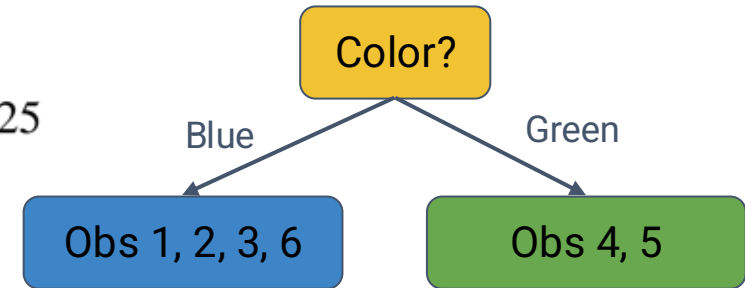
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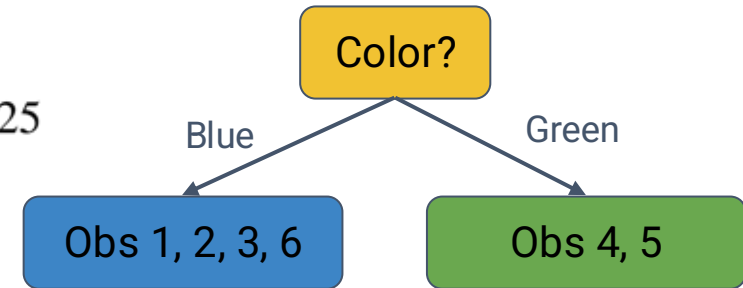
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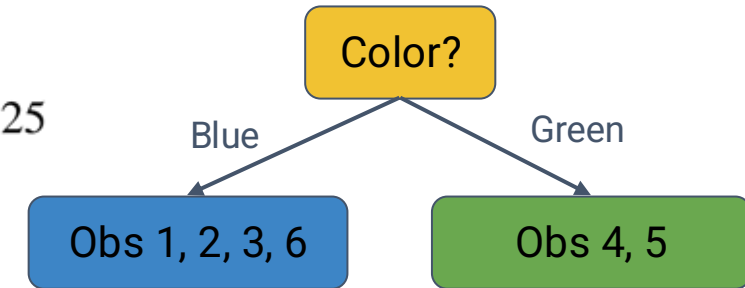
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$$B\left(\frac{3}{4}\right) = 0.81125$$



What attribute should we split on next for the Blue examples?

Compute the gain of splitting on **Spots**

$$\text{Remainder}(\text{Spots}) = \left(\frac{1}{4}\right)B\left(\frac{1}{1}\right) + \left(\frac{3}{4}\right)B\left(\frac{2}{3}\right)$$

$$\text{Remainder}(\text{Spots}) = 0.68875$$

$$\text{Gain}(\text{Spots}) = 0.81125 - 0.68875 = 0.12250$$

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Information Gain Example - SOLUTION

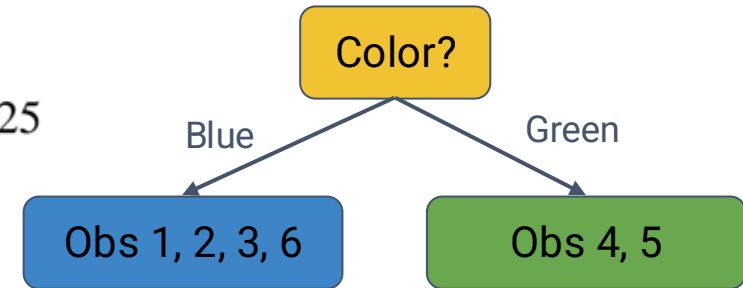
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$$B\left(\frac{3}{4}\right) = 0.81125$$



What attribute should we split on next for the Blue examples?

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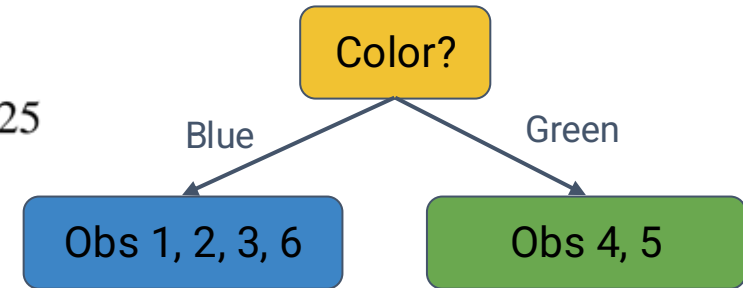
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Suppose we split on color first.

$$B\left(\frac{3}{4}\right) = 0.81125$$



What attribute should we split on next for the Blue examples?

Compute the gain of splitting on **Size**

$$\text{Gain}(\text{Spots}) = 0.12250$$

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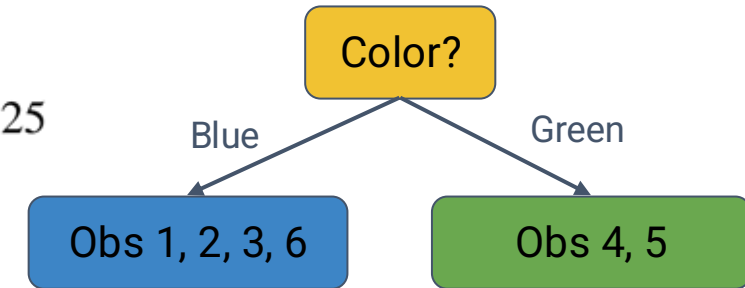
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Compute the gain of splitting on **Size**

$$\text{Gain}(\text{Spots}) = 0.12250$$

$$\begin{aligned} \text{Remainder}(\text{Size}) \\ = \left(\frac{3}{4}\right) B\left(\frac{3}{3}\right) + \left(\frac{1}{4}\right) B\left(\frac{0}{1}\right) \end{aligned}$$

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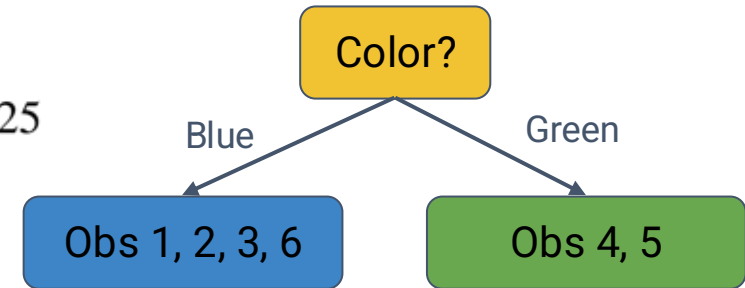
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Remaining
examples where
Size=Small

Entropy of
examples where
Size=Small

Observation	Size	Color	Spots	Poisonous
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2	Small	Blue	Yes	Yes
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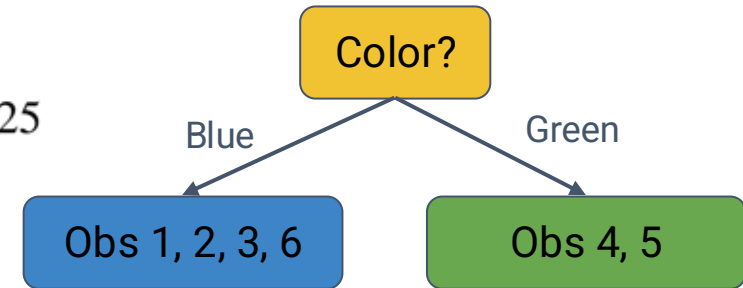
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Information Gain Example - SOLUTION

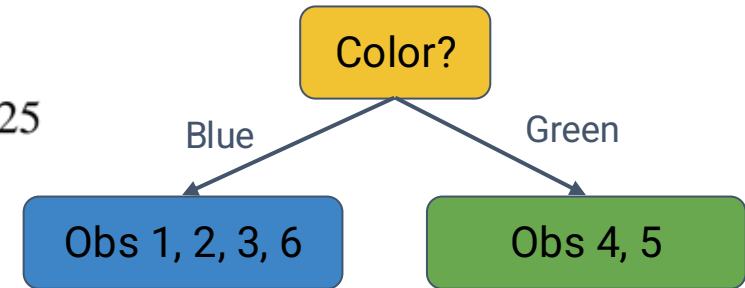
$$B(q) = -(q \log_2 q + (1 - q) \log_2 (1 - q))$$

$$\text{Remainder}(A) = \sum_{k=1}^d \frac{p_k + n_k}{p + n} B\left(\frac{p_k}{p_k + n_k}\right)$$

$$\text{Gain}(A) = B\left(\frac{p}{p + n}\right) - \text{Remainder}(A)$$

Suppose we split on color first.

$$B\left(\frac{3}{4}\right) = 0.81125$$



What attribute should we split on next for the Blue examples?

Compute the gain of splitting on **Size**

$$\text{Gain}(\text{Spots}) = 0.12250$$

$$\begin{aligned} \text{Remainder}(\text{Size}) &= \left(\frac{3}{4}\right) B\left(\frac{3}{3}\right) + \left(\frac{1}{4}\right) B\left(\frac{0}{1}\right) \\ \text{Remainder}(\text{Size}) &= 0 \end{aligned}$$

Observation	Size	Color	Spots	Poisonous
1	Small	Blue	No	Yes
2	Small	Blue	Yes	Yes
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4	Big	Green	No	No
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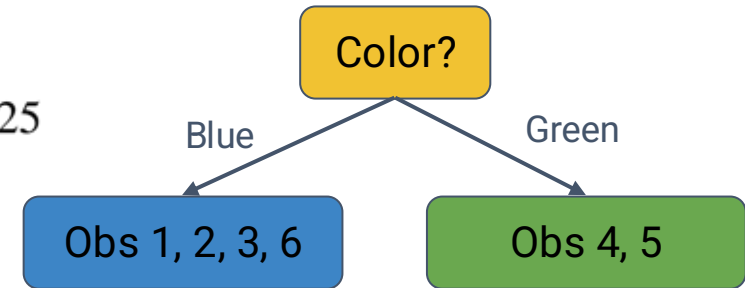
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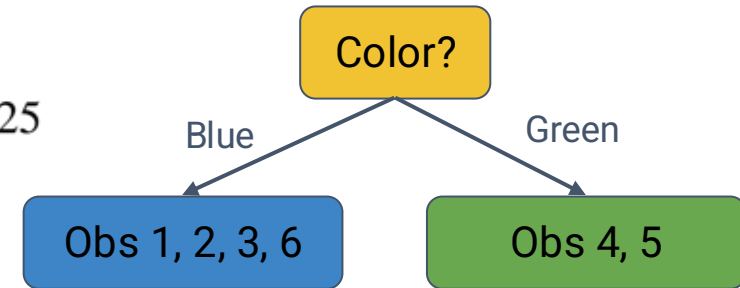
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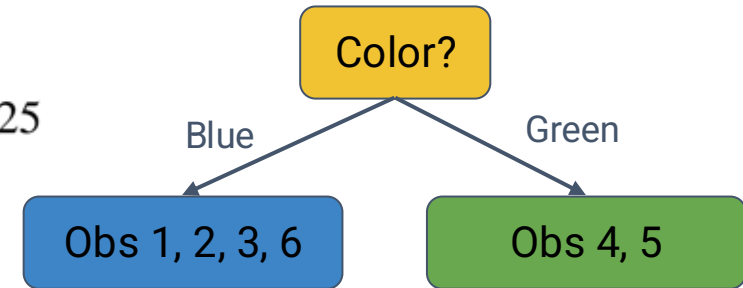
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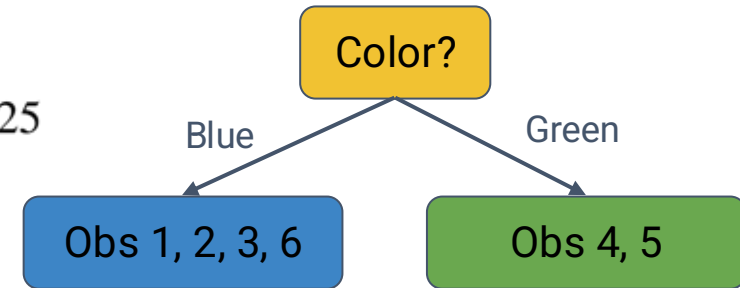
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Select the attribute with the greatest information gain

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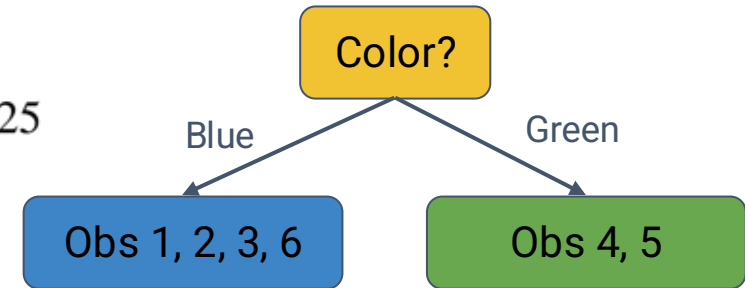
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Size completely classifies the remaining examples, so this is the best attribute!

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Ensembles: Bagging & Boosting

Combining many weak/varied trees into one strong predictor

Bagging (Bootstrap Aggregating)

- **Intuition**

- Train many trees in PARALLEL, each on a different random sample drawn WITH replacement (a “bootstrap”).
- Each tree is deep and high-variance — it overfits its own sample in its own way.
- Combine by majority vote (or averaging). Independent errors cancel out → reduces VARIANCE.

- **Example**

- Train 3 trees, each on a bootstrap sample of the 6 frogs, then vote on a new frog:

Tree	Obs drawn (with replacement)	Prediction
Tree 1	1, 1, 2, 3, 5, 6	Yes
Tree 2	2, 3, 3, 4, 5, 6	No
Tree 3	1, 2, 4, 5, 6, 6	Yes
Vote	Yes, No, Yes → majority	YES

Each tree saw slightly different data, so they make different mistakes — voting smooths them out.

Boosting

- **Intuition**

- Train weak trees (“stumps”) SEQUENTIALLY; each new one focuses on what the previous got wrong.
- Every example carries a weight. Misclassified points get their weight increased for the next round.
- Final answer = weighted vote; more accurate trees get a larger say. Reduces BIAS.

- **Example**

- 5 points, all weights start at 0.20. Stump h_1 misclassifies obs 3 and 5 → bump their weights:

Obs	Weight before	Correct?	Weight after
1	0.20	✓	0.125
2	0.20	✓	0.125
3	0.20	✗	0.25
4	0.20	✓	0.125
5	0.20	✗	0.25

Round 2 trains on the reweighted data, so h_2 works hard to get obs 3 & 5 right.

Final: $H(x) = \text{sign}(\alpha_1 h_1(x) + \alpha_2 h_2(x) + \dots)$, larger α for more accurate stumps.

Bagging vs. Boosting

- Both combine many trees — but train them very differently:

	Bagging	Boosting
Training	Parallel, independent	Sequential, dependent
Data per tree	Bootstrap sample	Reweighted full set
Tree strength	Strong (deep, overfit)	Weak (stumps)
Primarily reduces	Variance	Bias
Voting	Equal weight	Weighted by accuracy